

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	4
Total 1 mark			

Q2.

(a) $72 = 96p + q$

M1

$$24 = 24p + q$$

OE

M1

$$48 = 72p$$

Valid method to solve the correct two simultaneous eqns in p and q to at least the stage $48 = 72p$ OE

m1

$$p \left(= \frac{48}{72} \right) = \frac{2}{3}$$

AG CSO

A1

4

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(b) $q = 8$

Award if seen at any stage in this question

B1

$$u^3 = 48 + q \quad (u^3 =) 56$$

If not 56, ft on $(48 + c' q)$ provided at least M1 scored in part (a).

B1F

2

[6]

Q3.

(a) $48 = 60p + q$

$$12 = 12p + q$$

M1 for each equation in ACF

M1

(Condone embedded values for the M1M1)

M1

$$36 = 48p \text{ or } p = \frac{36}{48}$$

Valid method to solve the correct two simultaneous eqns in p and q to at least the stage $ap = b$ or $cq = d$

m1

$$p = \frac{3}{4}$$

AG (condone if left as equiv. decimal)

A1

$$q = 3$$

Can award if seen explicitly in (b) and no contradiction [ie not attempted in (a)]

B1

5

(b) $u_3 = 36 + q = 39$

If not 39, ft on $(36 + c's q)$

B1F

1

[6]

Q4.

(a) $u_2 = 6.8$

OE eg 34/5

B1

$$u_3 = 8.72$$

Ft on $6 + 0.4 \times c's u_2$

B1F

2

(b) $L = 6 + 0.4L$

Replacing u_{n+1} and u_n by L

M1

$$L = \frac{6}{1 - 0.4}$$

PI provided M scored

m1

$$L = 10$$

Must form an equation in L otherwise 0/3

A1

3

[5]

Q5.

(a) $24 = 16k + 12$

Condone with 0.75 (OE) subst for k

M1

$$k = 12 \div 16 = 0.75$$

AG; OE fraction; if verification must explicitly state the conclusion

A1

2

(b) $u_3 = 30$

B1

$$u_4 = 34.5$$

ft on $0.75 \times \text{cand's } u_3 + 12$

B1F

2

(c) (i) $L = 0.75L + 12$

Replacing u_{n+1} and u_n by L

M1

1

(ii)
$$L = \frac{12}{1 - k} = \frac{12}{1 - 0.75}$$

*PI, but previous M **must** be scored*

m1

$$L = 48$$

SC: (c)(i) incorrect and then in (c)(ii)

$$L = 0.75L + 12 \text{ leading to } L = 48 \text{ scores B2}$$

A1

2

[7]

Q6.

(a) $8 = -8p + q$

Either equation. PI eg by combined eqn.

M1

$$4 = 8p + q$$

Both (condone embedded values for the M1A1)

A1

Valid method to solve two simultaneous equations in p and q to find either p or q

m1

$$q = 6$$

AG (condone if left as a fraction)

A1

$$p = -0.25$$

OE

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B1

5

(b) $u_4 = 5$

Ft on $(6 + 4p)$

B1F

1

(c) (i) $L = pL + q;$ ($L = -0.25 L + 6$)

OE

M1

1

(ii)
$$L = \frac{q}{1-p}$$

Rearranging

m1

$$L = \frac{6}{1.25} = 4.8$$

$$Ft \text{ on } \frac{6}{1-p}$$

Dependent on previous two marks

A1F

2

[9]

Q7.

(a) $150 = 200p + q$

Either equation

M1

$$120 = 150p + q$$

Both (condone embedded values for the M1A1)

A1

*Valid method to solve two simultaneous eqns
in p and q to find either p or q*

m1

$$p = 0.6$$

AG (condone if left as a fraction)

A1

$$q = 30$$

B1

5

(b) $u_4 = 102$

Ft on $(72 + q)$

B1F ✓

1

(c) $L = pL + q; \quad L = 0.6L + 30$

M1

$$L = \frac{q}{1-p}$$

m1

$$L = 75$$

Ft on 2.5q

A1F ✓

3

[9]